CS253 — Data Structures and Algorithms for DS/AI

PSO 6 • Week 7 • Hash Tables & Runtime Comparisons

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Spring 2025 (style) • Student-facing HTML version

Notes: Unless stated otherwise, assume standard word-RAM conventions. For hashing questions, follow the assumptions stated in each problem.

Question 1 — Double Hashing Insert Trace

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Let $S = \langle 91,\ 44,\ 130,\ 57,\ 26,\ 71,\ 148,\ 102\rangle$ be a sequence of keys. Insert the keys into a hash table of size $m = 17$ using **open addressing** with **double hashing**:

$$h_1(k) = k \bmod 17$$
$$h_2(k) = 1 + (k \bmod 13)$$

Probe sequence: $h(k,i) = (h_1(k) + i \cdot h_2(k)) \bmod 17$ for $i = 0, 1, 2, \ldots$

Show the probe sequence for each key until insertion succeeds, and write the final table contents (index $\to$ key).

Tip: Since m is prime and $h_2(k) \in \{1, \ldots, 13\}$, you should get full-cycle probing.

Question 2 — Worst-Case Runtime Table

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Fill in the table with the **asymptotic worst-case** runtime for each operation. For hash tables, assume **independent and uniform hashing (I.U.H.)** for the *hashing* part, but remember the question asks for **worst-case**.

GetSuccessor(x) takes a reference to an element already stored in the structure and returns the next larger key.

Operation	Hash Table (Separate Chaining)	Hash Table (Open Addressing)	BST (Unbalanced)	Balanced BST
Insert	________	________	________	________
Search	________	________	________	________
Delete	________	________	________	________
GetMax	________	________	________	________
GetSuccessor(x)	________	________	________	________

Hint: For BST successor, think about parent pointers vs. walking from root; state your assumption briefly.

Question 3 — Expected Probes (Short Proof)

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Assume **open addressing** with independent and uniform hashing. Let $\alpha = n/m$ be the load factor, where n keys are stored in a table of size m, and assume $\alpha < 1$.

Prove that the expected number of probes in an **unsuccessful** search is:

$$\mathbb{E}[\text{probes}] = \frac{1}{1-\alpha}$$

Suggested approach: express the expected probes as a sum of probabilities that the first $t-1$ probes hit occupied slots. Use the fact that under I.U.H., each probe is uniformly distributed over table positions.
